

CareFlow AI

Healthcare Operations and Post-Discharge Follow-Up Platform

Architecture Documentation

Queue-Driven		AI-Assisted
Multi-Tenant		Human-in-the-Loop
Observable		Safety-Conscious

Document:	System Architecture
Version:	1.0
Environment:	Prototype / Demonstration
Primary Workflow:	Post-discharge patient follow-up

This document describes the architecture, responsibilities, data boundaries, workflow, AI safety boundaries, queue processing, and prototype trade-offs of CareFlow AI.

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1 Executive Summary

CareFlow AI is a multi-tenant healthcare operations platform designed to coordinate post-discharge patient follow-up.

The system connects patient and discharge data, campaign management, eligibility evaluation, centralized outreach queues, patient responses, AI-assisted clinical triage, hospital-specific knowledge retrieval, human escalation workflows, controlled EHR operations, notifications, audit logging, dashboards, and operational observability.

The architecture intentionally treats AI as one controlled subsystem inside a larger operational workflow rather than implementing the entire product as one unrestricted AI agent.

The central design principle is:

**Patient Data → Eligibility → Queue → Outreach → AI Assessment → Consensus →
Human Review / Controlled Completion**

The prototype also contains a deterministic queue simulation. The simulation reuses the same seeded demo patients and outreach tasks rather than creating a separate simulation patient population.

2 Architectural Goals

The architecture is designed around the following goals.

2.1 Operational Goals

- Support multiple hospitals as isolated tenants.
- Manage hospital users using role-based access control.
- Import and manage structured patient and discharge information.
- Create campaigns and evaluate patient eligibility.
- Place eligible patients into a centralized outreach queue.
- Enforce hospital-level outbound capacity.
- Support priority-based queue processing.
- Support retries, callbacks, dropped interactions, and maximum attempts.
- Track clinical deadlines and operational state.

2.2 AI Goals

- Ground AI processing in hospital-specific protocols and knowledge.
- Convert patient responses into structured clinical information.
- Perform multiple independent assessment passes.
- Compare assessment results through a consensus stage.
- Detect disagreement rather than silently hiding it.
- Escalate cases requiring human attention.
- Keep AI actions behind controlled backend tools and business rules.

2.3 Safety Goals

- Prevent automatic completion of cases requiring human review.
- Enforce tenant-aware retrieval.
- Validate structured AI outputs before application use.
- Keep audit records for important operations.
- Prevent duplicate external side effects through idempotent operations.
- Make queue and AI behavior observable.

3 System Context

CareFlow AI sits between hospital operational data and patient follow-up interactions.

The system receives patient and discharge information, determines which patients require outreach, schedules and processes outreach, collects responses, and routes those responses through a controlled AI and human-review workflow.

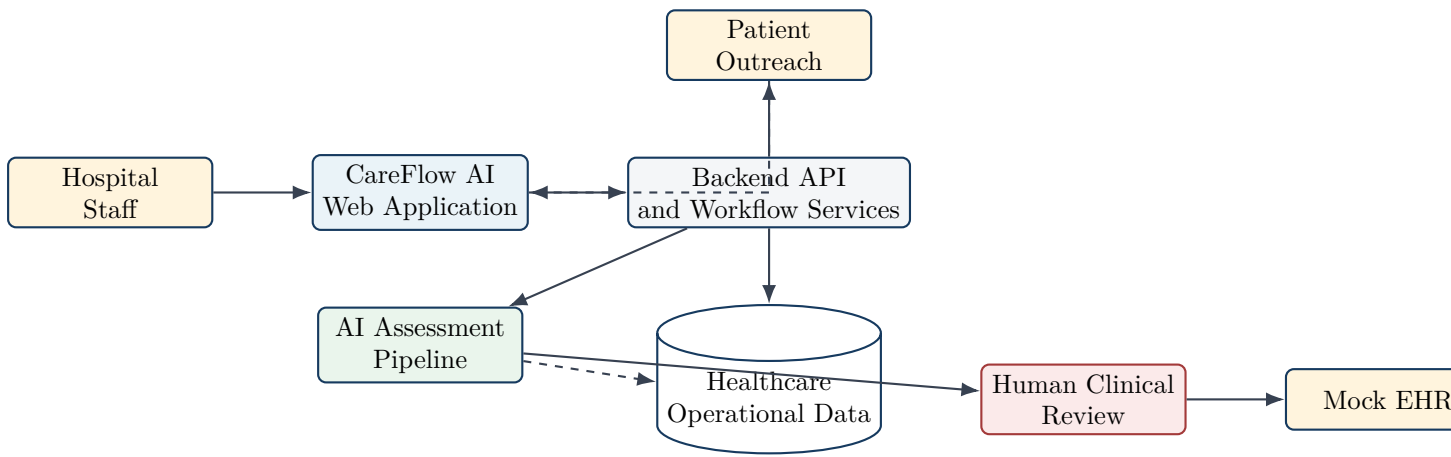


Figure 1: CareFlow AI system context

4 High-Level System Architecture

The system is organized into several logical layers.

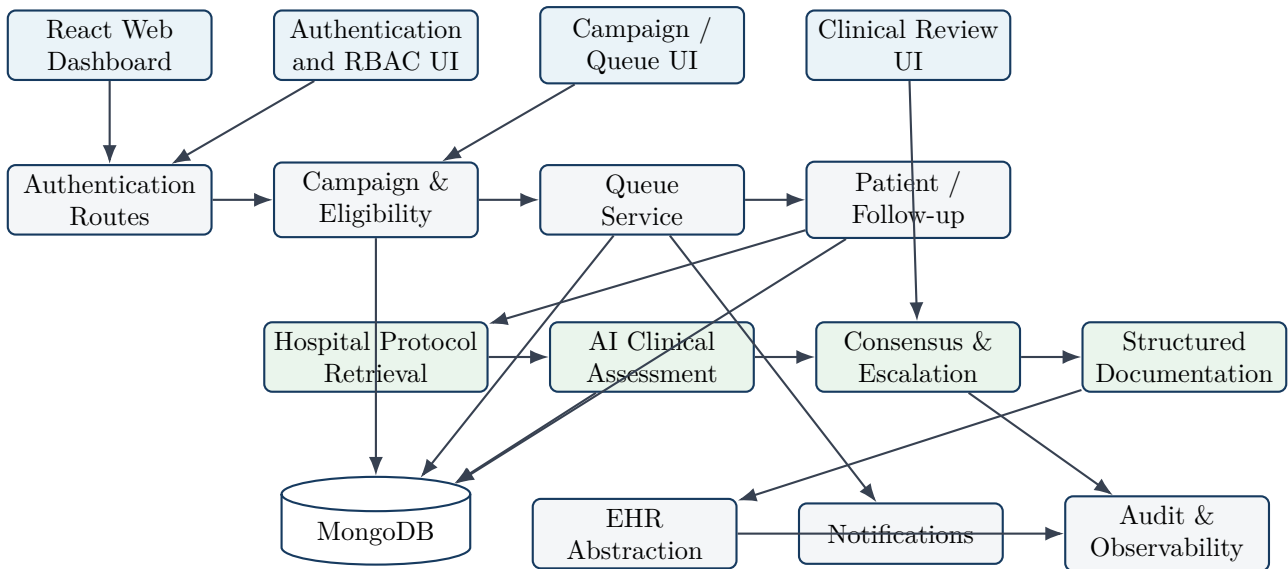


Figure 2: Logical architecture of CareFlow AI

5 Major Components and Responsibilities

5.1 Frontend

The frontend provides the operational interface for authorized users.

Major views include:

- Authentication and role-aware access.
- Hospital dashboard.
- Hospital configuration.
- Patient operational view.
- Campaign management.
- Eligibility processing.
- Outreach queue.
- Clinical review.
- Queue simulation.

The frontend communicates with the backend through HTTP APIs. Business rules are kept in the backend rather than being trusted to the browser.

5.2 Authentication and Authorization

The authentication layer provides:

- User authentication.
- Role-based access control.
- Hospital/tenant context.
- Protected API access.

Authorization is applied before protected business operations.

The architecture supports roles such as platform-level and hospital-level users, with permissions determined by the user's authorized scope.

5.3 Hospital and Tenant Management

Each hospital represents an isolated tenant.

Hospital configuration includes operational information, capacity constraints, escalation contacts, and hospital-specific knowledge/protocol resources.

Tenant context is propagated through API operations and background processing so that a patient from one hospital does not retrieve another hospital's protocol.

5.4 Patient and Healthcare Data

Patient information is represented using healthcare-oriented entities rather than a single flat patient object.

The data model supports concepts such as:

- Patient
- Encounter
- Condition
- CarePlan
- Discharge information
- Outreach Task
- Communication
- Escalation
- Audit record

This provides a foundation for future FHIR-shaped integrations while keeping the prototype implementation manageable.

6 Campaign and Eligibility Architecture

Campaigns determine which patients require follow-up.

The eligibility workflow is:

1. Campaign is created or activated.
2. Patient population is evaluated.
3. Discharge and follow-up criteria are checked.
4. Eligible patients receive outreach tasks.
5. Each task is associated with its hospital and campaign.
6. Priority and deadline information are calculated.
7. Tasks enter the centralized queue.

For the demonstration environment, the database is seeded with 30 demo patients across two demo hospitals.

The same demo patients are reused by the deterministic queue simulation.

7 Queue and Scheduling Architecture

The queue is a central operational component.

It determines which eligible patient should be processed next while respecting:

- Hospital capacity.
- Task priority.
- Clinical deadlines.
- Retry state.
- Callback state.
- Task locks.
- Failure recovery.

7.1 Queue State Model

A simplified task lifecycle is:

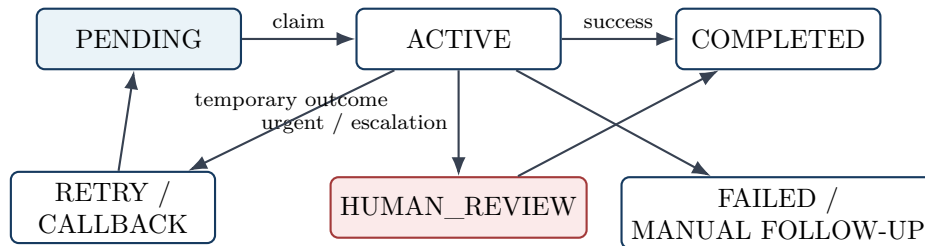


Figure 3: Simplified outreach task lifecycle

The actual task contains additional operational information such as attempt count, lock state, callback information, deadlines, and simulation metadata where applicable.

8 Concurrency and Capacity Control

Outbound processing is controlled at the hospital level.

For each hospital, the queue maintains a relationship between:

$$\text{activeOutboundCount} \leq \text{configuredCapacity}$$

Before a task is claimed, the worker attempts to reserve an available capacity slot.

A successful reservation allows processing to continue.

If capacity is full, the task remains queued.

This prevents workers from independently exceeding the hospital's configured outbound concurrency.

The prototype demonstrates this behavior using constrained capacities for the demo hospitals.

9 Outreach Architecture

The outreach layer is responsible for executing the communication selected by the queue.

The prototype supports controlled email outreach for the demonstration.

The real outreach path is:

Queue Claim → Capacity Reservation → Outreach Preparation → Notification Delivery →
Communication Record → Capacity Release

Important operations are logged and recorded for traceability.

The demonstration environment restricts patient outreach to controlled test inboxes.

10 AI Architecture

The AI subsystem is intentionally separated into logical responsibilities instead of using one unrestricted agent.

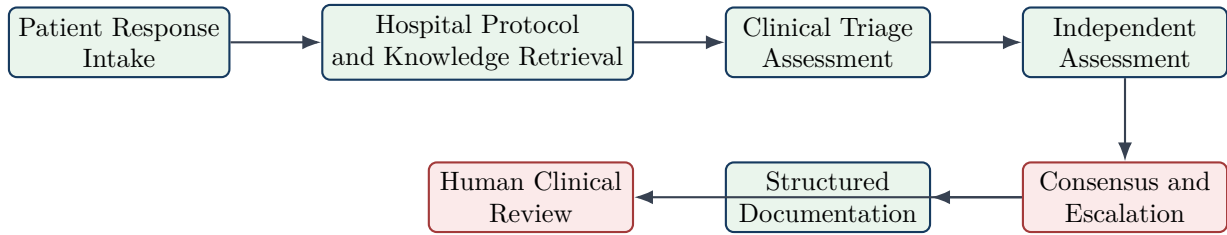


Figure 4: AI processing pipeline

10.1 Patient Response Intake

The patient follow-up interface can collect structured information from the patient.

The prototype supports text-based responses and may support voice input depending on the configured demonstration workflow.

The response becomes input to downstream processing.

10.2 Hospital-Grounded Retrieval

The AI processing retrieves only the hospital-specific protocol and knowledge relevant to the current task.

This is a critical tenant boundary:

$$Patient_{HospitalA} \not\rightarrow Knowledge_{HospitalB}$$

The retrieval result retains protocol or source context so that the downstream assessment can be traced back to approved knowledge.

10.3 Clinical Triage

The triage stage converts the patient interaction into structured information.

A conceptual output contains:

- Classification: routine, concerning, urgent, or uncertain.
- Observed indicators.
- Evidence from the interaction.
- Relevant protocol references.
- Confidence or uncertainty information.
- Escalation recommendation.

AI output must be validated before application logic consumes it.

11 Consensus and Escalation

Escalation is treated as a safety-sensitive workflow.

Rather than relying exclusively on a single assessment, CareFlow performs multiple assessment passes and compares their results.

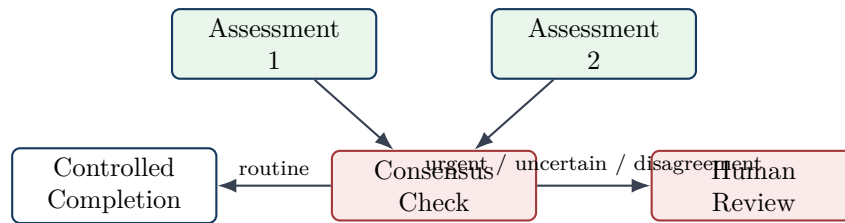


Figure 5: Consensus and escalation boundary

The consensus layer identifies agreement and disagreement.

A disagreement between assessment paths is not silently discarded. Instead, it becomes part of the safety workflow and can trigger conservative escalation.

For an urgent case, the resulting state is:

HUMAN_REVIEW

rather than automatic completion.

12 Human-in-the-Loop Clinical Workflow

When escalation occurs, the system creates a clinical review task.

The workflow is:

1. AI assessment identifies a case requiring attention.
2. Escalation record is created.
3. Outreach task moves to **HUMAN_REVIEW**.
4. Authorized staff can view the case.
5. Reviewer assigns or accepts the escalation.
6. Reviewer acknowledges the case.
7. Reviewer examines the available structured information.
8. Reviewer resolves the escalation.
9. Final workflow state is recorded.

This boundary prevents an urgent AI classification from being treated as an unrestricted automatic clinical action.

13 Mock EHR and Controlled Tools

CareFlow contains an EHR abstraction layer.

The AI layer does not directly manipulate database tables.

Instead, an AI-driven healthcare operation follows:

AI Decision → Structured Tool Request → Authorization → Schema Validation → Business Rule
Validation → EHR Operation

The prototype uses a mock EHR implementation.

This abstraction allows a future implementation to replace the mock adapter with a real EHR integration without coupling AI logic directly to the external EHR.

EHR operations are observable and failures are represented explicitly.

14 Data Model

The logical data model is organized around hospital tenancy and patient workflow.

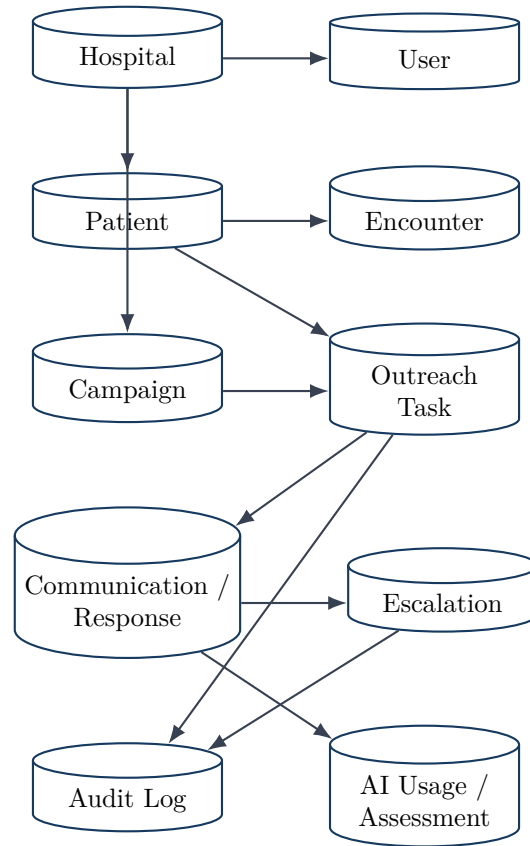


Figure 6: Logical healthcare workflow data model

Important design properties include:

- Hospital identifiers provide tenant boundaries.
- Patient records are associated with encounters and care plans.
- Campaigns produce outreach tasks.
- Outreach tasks maintain operational state.
- Communications capture outreach and patient interaction.
- Escalations represent human-review workflows.
- Audit records capture important state-changing operations.
- AI assessment information provides traceability for AI-driven decisions.

15 Notifications and Background Workflows

Notifications are separated from the core queue and clinical workflow logic.

The notification service handles delivery attempts and reports success or failure.

Conceptually:

$$WorkflowEvent \rightarrow NotificationService \rightarrow ExternalChannel$$

Failures are represented explicitly rather than assuming external delivery always succeeds.

Background workflows can therefore support:

- Retry processing.
- Callback scheduling.
- Escalation notifications.
- Failed-operation recovery.
- Queue maintenance.

16 Auditability and Observability

CareFlow makes important operations observable.

Examples include:

- Authentication events.
- Campaign lifecycle actions.
- Eligibility processing.
- Queue claims.
- Capacity reservation.
- Outreach delivery.
- AI assessment.
- Consensus decisions.
- Escalation creation.
- Human reviewer actions.
- EHR operations.
- Simulation operations.

The prototype also emits structured operational logs for major workflow transitions.

Examples include:

```
[ELIGIBILITY] Starting eligibility
[QUEUE] Task atomically claimed
[OUTREACH] Preparing email outreach
[AI] Assessment #1 completed
[CONSENSUS] Result
[ESCALATION] Creating clinical review
[REVIEW] Escalation acknowledged
[EHR] Mock EHR update complete
[AUDIT] action | entity:id | hospital=...
```

Sensitive healthcare information should not be unnecessarily exposed in application logs.

17 Reliability, Idempotency and Failure Recovery

The architecture treats external operations as failure-prone.

Relevant failure classes include:

- Notification delivery failure.
- AI provider failure.

- Malformed AI output.
- Database failure.
- EHR operation failure.
- Queue worker failure.
- Timeout.
- Rate limiting.
- Interrupted outreach.

The system uses explicit task state, retry behavior, and controlled recovery paths. Operations with potentially harmful duplicate side effects should be idempotent. Examples include:

- Creating an escalation.
- Sending an outreach message.
- Scheduling a retry.
- Writing an EHR record.
- Recording an interaction outcome.

A failed worker should not permanently consume hospital outbound capacity.

18 Security and Privacy Boundaries

The prototype demonstrates healthcare-data security awareness through:

- Authentication.
- Role-based authorization.
- Tenant isolation.
- Input validation.
- Controlled API access.
- Secret management through environment configuration.
- Audit logging.
- Data minimization.
- Controlled AI tool access.

18.1 AI Security Boundary

Patient-provided content and retrieved documents are treated as untrusted input.

They must not be able to override:

- System instructions.
- Authorization rules.
- Hospital safety policies.
- Tool access restrictions.

The prototype should not be represented as a production-certified healthcare system. Production deployment would require additional security, compliance, privacy, infrastructure, monitoring, and clinical validation controls.

19 Multi-Tenancy

Tenant isolation is a cross-cutting architectural requirement.

The hospital identifier is carried through:

API → Business Logic → Database → Queue → AI Retrieval → Tools / EHR → Analytics

Every protected operation should verify that the authenticated user has access to the target hospital.

Hospital-specific knowledge retrieval is also tenant-aware.

Therefore:

$$Tenant(A) \neq Tenant(B)$$

and information belonging to Hospital A must not be used to answer a Hospital B workflow.

20 Queue Simulation Architecture

The queue simulation is a deterministic operational demonstration layer.

It does not create an independent simulation patient population.

Instead:

Seeded Demo Patients → Campaign Eligibility → Existing Outreach Tasks → Simulation

The simulation reuses the existing outreach tasks and attaches deterministic outcome scenarios to them.

Example outcomes include:

- Completed outreach.
- No answer.
- Busy.
- Voicemail.
- Dropped interaction.
- Callback.
- Retry.
- Maximum retry condition.
- Escalation.

This makes the demonstration repeatable without requiring paid telephony infrastructure.

21 End-to-End Workflow

The complete workflow is illustrated below.

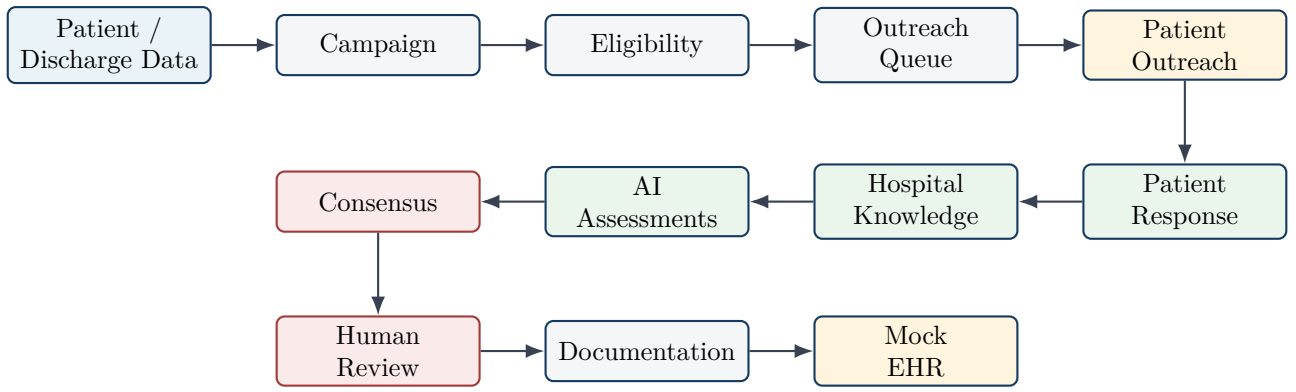


Figure 7: End-to-end CareFlow AI workflow

The operational sequence is:

1. Hospital and users are configured.
2. Patient and discharge data are available.
3. A campaign is created or activated.
4. Eligibility identifies patients requiring outreach.
5. Outreach tasks enter the centralized queue.
6. Queue capacity and priority determine processing.
7. Controlled outreach is executed.
8. The patient provides a response.
9. Hospital-specific knowledge is retrieved.
10. Multiple AI assessments evaluate the response.
11. Consensus compares the assessments.
12. Routine cases follow the controlled completion path.
13. Urgent, uncertain, or escalated cases enter human review.
14. Documentation and controlled EHR operations are performed where permitted.
15. Audit and operational logs capture important events.
16. Dashboards expose workflow and system state.

22 Prototype Deployment Architecture

The prototype is structured as a conventional web application:

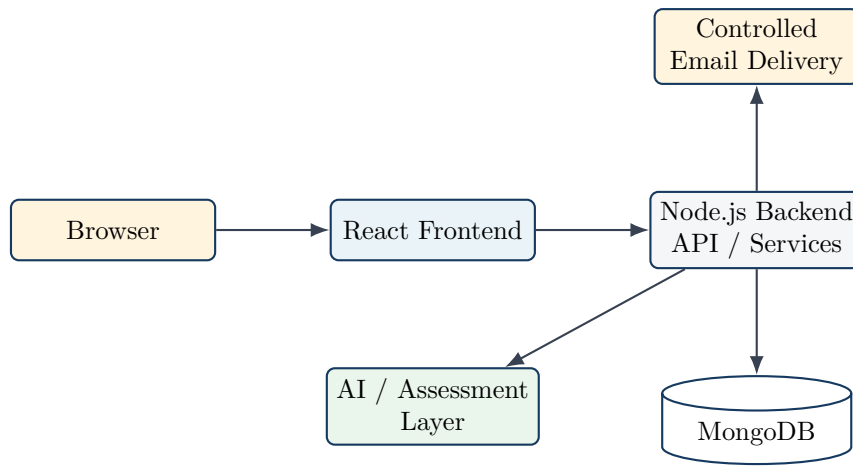


Figure 8: Prototype deployment view

Environment-specific configuration and secrets are supplied through environment variables rather than committed into source control.

23 Scalability Considerations

The prototype architecture can be extended for larger workloads.

23.1 Horizontal Worker Scaling

Queue workers can be scaled horizontally while maintaining centralized hospital capacity control.

23.2 Database Scaling

Indexes should be maintained on frequently queried fields such as:

- Hospital identifier.
- Campaign identifier.
- Task status.
- Priority.
- Scheduled time.
- Clinical cutoff.
- Lock state.

23.3 AI Scaling

AI calls are latency-sensitive and can be moved into asynchronous workers.

The application should track AI request state independently from the HTTP request lifecycle.

23.4 Event Processing

As workload grows, queue events, notifications, and AI jobs can be moved to a dedicated message broker or distributed job system.

24 Technology Mapping

The current prototype maps the logical architecture to the following technology categories.

Layer	Prototype Responsibility
Frontend	React-based web application for hospital operations, campaigns, patients, queues, reviews, and simulation.
Backend	Node.js backend exposing APIs and coordinating business workflows.
Data	MongoDB-based operational persistence for hospitals, patients, campaigns, tasks, communications, escalations, and audit data.
Queue	Centralized task processing with priority, capacity, retry, callback, and state management.
AI	Structured assessment pipeline with hospital-grounded retrieval, multiple assessment passes, consensus, and escalation.
EHR	Mock EHR abstraction providing controlled healthcare record operations.
Notifications	Controlled email notification and outreach delivery.
Observability	Operational logs, audit records, workflow state, and system health information.
Simulation	Deterministic queue simulation using the existing demo task dataset.

Table 1: Logical architecture to implementation mapping

25 Prototype Trade-offs

CareFlow is intentionally scoped as a prototype.

The following trade-offs are explicit:

- The EHR is mocked rather than connected to a production EHR.
- Demonstration outreach uses controlled email rather than requiring paid production telephony.
- Queue simulation uses deterministic scenarios for repeatability.
- The healthcare data representation is simplified while retaining healthcare-oriented entities.
- AI functionality is structured around the demonstrated assessment and consensus workflow.
- Production-scale infrastructure, compliance certification, and clinical deployment controls are outside the prototype scope.

These are intentional prototype boundaries rather than assumptions that the demonstration environment is production-ready.

26 Critical Safety Boundaries

The most important architectural safety boundaries are:

1. **Tenant boundary:** Hospital-specific data and knowledge remain isolated.
2. **Queue boundary:** Outbound capacity is centrally controlled.
3. **AI boundary:** AI produces structured outputs that are validated by the backend.
4. **Knowledge boundary:** AI retrieval is constrained to the relevant hospital context.
5. **Consensus boundary:** Multiple assessments are compared and disagreement is visible.
6. **Human boundary:** Urgent or uncertain cases can enter human review.

7. **Tool boundary:** AI does not directly manipulate healthcare database tables.
8. **Audit boundary:** Important state-changing operations are recorded.
9. **External-system boundary:** EHR and notification operations are isolated behind service abstractions.

27 Conclusion

CareFlow AI is architected as a connected healthcare operations platform rather than an isolated AI chatbot.

Its core architecture connects:

**Multi-Tenancy → Patient Data → Campaigns → Eligibility → Queue → Outreach → AI
→ Consensus → Human Review → EHR**

The architecture separates responsibilities so that queue management, clinical AI processing, healthcare data, EHR operations, notifications, and observability can evolve independently.

The most safety-sensitive boundary is the transition from AI assessment to clinical action. AI assessment is therefore followed by structured validation, multiple assessments, consensus, and human review for cases requiring escalation.

The prototype's deterministic simulation reuses the same seeded demo patients and tasks, allowing queue capacity, retries, callbacks, deadlines, dropped interactions, and escalation behavior to be demonstrated repeatedly without introducing a second patient dataset.

Overall, the architecture provides a foundation for extending the prototype toward asynchronous workers, richer AI observability, production-grade healthcare integrations, advanced analytics, and larger multi-hospital deployments.